Converting Product to Summation

$$S_n = \prod_{k=1}^{n} a_k \quad (a_k > 0)$$

$$\ln(S_n) = \sum_{k=1}^{n} \ln(a_k)$$

Since $\ln x$ is continuous on $(0, \infty)$,

$$\ln\left(\prod_{k=1}^{\infty} a_k\right) = \ln\left(\lim_{n\to\infty} P_n\right) = \lim_{n\to\infty} \ln(P_n)$$

According to logarithm rule:

$$\ln(a_0 \times a_1 \times a_2 \times \cdots \times a_n) = \ln(a_0) + \ln(a_1) + \ln(a_2) + \cdots + \ln(a_n)$$

Thus,

$$\ln(S_n) = \ln\left(\prod_{k=1}^{n} a_k\right) = \sum_{k=1}^{n} \ln(a_k)$$